

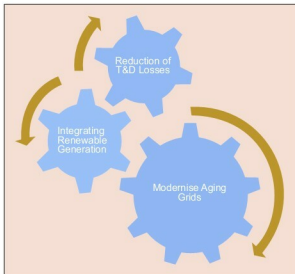
tune of 200km. In urban areas, meters are placed in very close proximity and the environment is not 100 per cent stable for this. Thakurdesai explains that the environment keeps changing, and finding the technology that works in this kind of environment is pretty challenging. Additionally, there are 50+ standards in the electrical meter segment. These vary based on the utility providers, and every state comes out with its own set of features and standards. These problems exist at the product level but these challenges have to be worked hand-in-hand with the consumers.

Lack of communication standards.

There are no standard protocols for communication in India. Various utilities experiment in a different manner. Maharashtra has looked at Zigbee protocol for AMR-related data collection. "Certain states are using proprietary 2.4GHz for communication; we also have seen states like Madhya Pradesh and Andhra Pradesh with sub-1G requirements. For the frequency band of 865MHz to 867MHz, the communication topology or protocol are still not standardised," adds Thakurdesai.

New consumption models. The new consumption models that have come up, such as automated buildings and electric vehicles, can be utilised to balance the stress on the grid. Systems should be in place to ensure that the energy storage and control solutions built into these can be used. Such as ensuring that they are set into consumption mode when the stress on the grid is low, and into energy source mode (if available through PV or energy storage) when the grid is in its peak load time.

Securing power supply. Electricity demand is increasing significantly, especially when basic power supply is extended to those remote areas that currently lack it. In this scenario, smart grid technologies should be able to help with efficient and reliable transmission and distribution (T&D) of electricity. The current SCADA-based system will no more be opened up to access once smart grid technologies



What needs to be done to get things moving

are integrated to it, so the challenge of keeping it secure becomes a bigger one.

Labour unions. A potential area for trouble could come from labour unions. The implementation of smart grid technologies brings about the same kind of problems that you would expect from automating anything—a lot of manual labour gets converted into skilled labour. As Sandra Diethelm, CEO, Exuta (India) puts it, skill sets will change from climbing power poles to data analysts sitting in offices.

Other challenges for implementing smart grid in the Indian scenario include access to electricity for 80 million households, availability of quality power at least during evenings, and reducing losses at the transmission and distribution side.

What can we expect as smart grid technologies get implemented

First, let us see why electricity suddenly becomes very important in the scenario. "This is because electricity is becoming a scarce product," explains Srinivasan. He adds that what was earlier a capital investment has now sort of become a product investment. This makes people cautious with respect to using electricity.

As of now with respect to telecommunication, if you do not pay your phone bill, they disconnect your

connection. Likewise, there is going to be control for electricity in the near future, i.e., utility can control your power. If I do not pay my electricity bill, my power will be cut off. Of course there will be other parameters that will be considered for cutting electricity supply, like if you are overshooting your margin demand (MD) control, or if you are introducing some harmonics, etc. These things will be monitored by the utility

and the end user will also be cautioned. There will be total control, and this is where we are heading.

Srinivasan says that one thing that has already started on a larger scale in India is stock exchanges for power. One person will be bidding for the power for a particular region for a particular period. "Suppose, I want to bid for Bangalore for a particular period of time, I will specify how much power I might need and bid for it. If successful, I will be sanctioned the specified amount of power for that amount and I can then sell it."

For this facility, there is a national grid element called load dispatch centre (LDC), which gets information from all generators who are pumping power into the grid. One will be aware of the power available, on a national level, for distribution and also how many people are consuming it. For this purpose, either interface meters or smart grid meters are used. From the generating station, the LDC will be measuring how much power is being pumped in online, power quality and status of the breakers, i.e., if they are switched on or off. There are various levels of hierarchy for LDC operations such as national, regional, state level and sub-load dispatch centers, adds Srinivasan. ●

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